Féidearthachtaí as Cuimse
Infinite Possibilities

Secure Communications & Cryptography (MACS H6014)

Symmetric Encryption

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Security by Obscurity

- "***Security by obscurity***" is a flawed approach to protecting systems or data.
- Relying on the secrecy or obscurity of certain aspects.
- Attempting to hide vulnerabilities or weaknesses in a system through proper security measures.
- Most of our classic ciphers relied on keeping the algorithm secret.
- ***What happens if the enemy figures out the algorithm? How can we fix it?***
- Much better to use a key-based algorithm. We just need to change the key, not the algorithm!
- All modern encryption is completely open, with all parts of the implementations publicly available.
- But criminals use cryptography, so we should create backdoors or at least control the algorithms?



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Cryptography Mechanism

- Fundamental building blocks of cryptography.
- Used for data protection, authentication, and secure communication.
- Mechanisms are mathematical techniques and algorithms.
- Cryptography is more than just encryption and decryption ciphers.
  - Integrity – Hashing
  - Authentication – Digital Certificates, PKI
  - Key exchange – Asymmetric encryption
  - Confidentially – Symmetric encryption
  - Anonymity, Digital Cash, E-Voting, Blockchain, Crypto-Currency, etc…

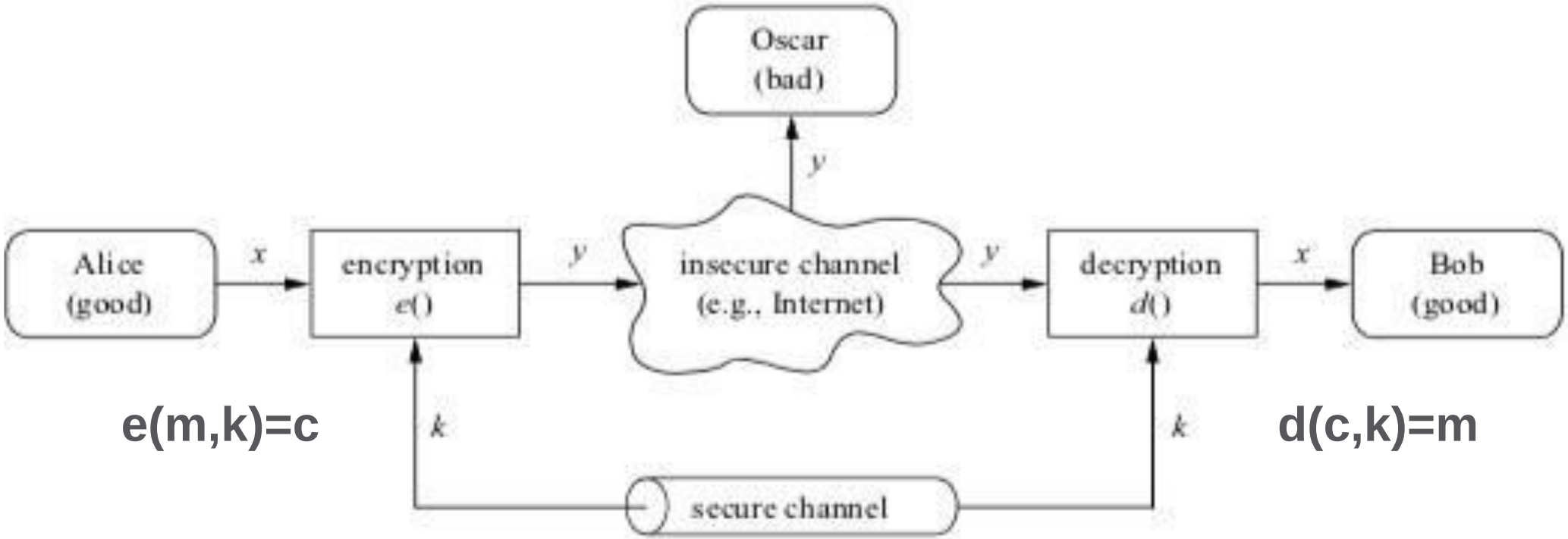


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Key-based Cryptography

- *m* is called ***plaintext*** or ***cleartext***
- *y* is called ***ciphertext***
- *k* is called the ***key***
- the set of all possible keys is called the ***key space***



e(m,k)=c

d(c,k)=m

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Some Points About Keys

- Terms that are used to describe keys are:
  - Key length is measured in bits.
  - Keyspace This is the number of possibilities that can be generated by a specific key length.
- **For example, if the key length is 2, then keyspace is 00, 01, 10, 11 = 2^2 = 4.**
- **For example, if key length is 3, then keyspace is 000, 001, 010, 011, 100, 101, 110, 111 = 2^3 = 8.**
- As key length increases, the keyspace increases exponentially.
- In the DES block cipher, a 56-bit key is used, resulting in a relatively small key space of size 2^{56} (or 7.2058 x 10^{16} combinations of characters), which can be easily searched with a standard desktop computer.

DES Key	Keyspace	# of Possible Keys
56-bit	2^{56} 111111 111111 111111 111111 111111 111111 111111 111111	72,000,000,000,000,000
57-bit	2^{57} 111111 111111 111111 111111 111111 111111 111111 111111 1	144,000,000,000,000,000
58-bit	2^{58} 111111 111111 111111 111111 111111 111111 111111 111111 11	288,000,000,000,000,000
59-bit	2^{59} 111111 111111 111111 111111 111111 111111 111111 111111 111	576,000,000,000,000,000
60-bit	6^{60} 111111 111111 111111 111111 111111 111111 111111 111111 1111	1,152,000,000,000,000,000

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XOR-Function Example

- Let's assume our message is m='CS'. We have to convert the string to an element of $M:=\{0,1\}^n$ where n= 7. That means every character in every message is represented by 7 bits.

'CS' => [1, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1]

and it follows with a **random key k**.

$$
\begin{aligned}
m = \text{'CS'} &= \overbrace{1010011}^{\text{C}}\overbrace{1000011}^{\text{S}} \\
&\quad \oplus \\
k &= 11001000100110 \\
m \oplus k = c &= 01101111100101
\end{aligned}
$$

If someone got the ciphertext but not the key, then it is not possible to compute the message however if you had the plaintext and ciphertext, then you could work out the k value.

Exercise: What would be the key to decrypt with a result is ' instead of the correct string '.

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Types of Encryption

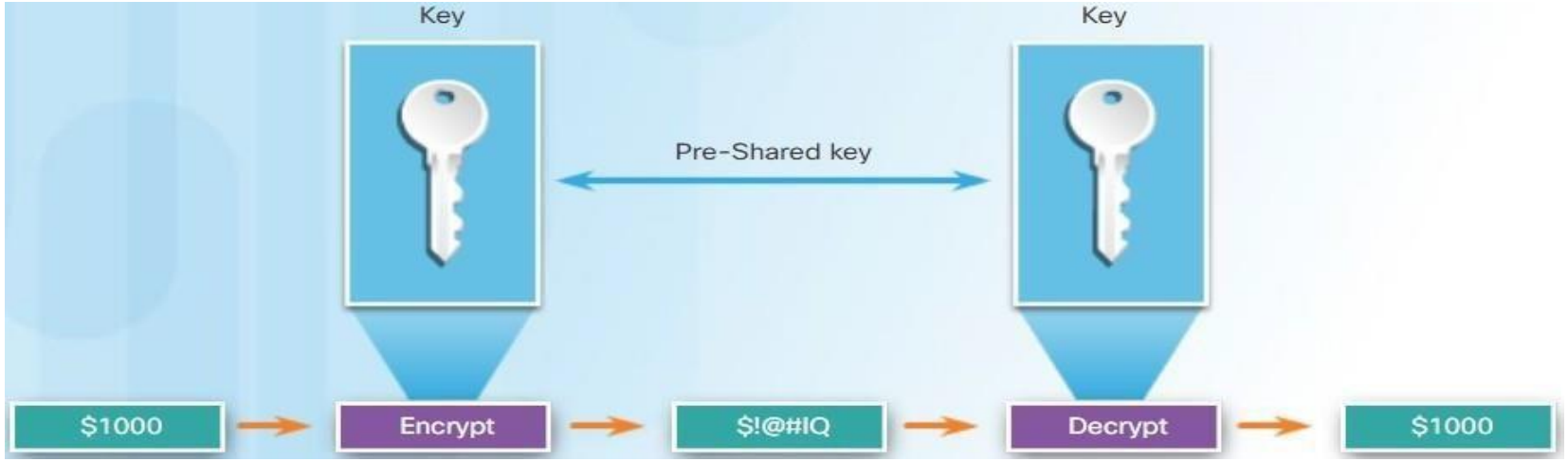
Symmetric	Asymmetric
Same Key	Different Key (Public / Private)
Extremely Fast	Relatively Slow
Smaller keys (128/256/512)	Bigger keys (2048, 4096, 8192+)
Key exchange is a problem	Good at key exchange
Only offers Confidentiality	Can also offer Authentication as well as confidentiality
3-DES, RC4, RC5, AES	DH, RSA, ElGamal, DSA

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Symmetric Encryption

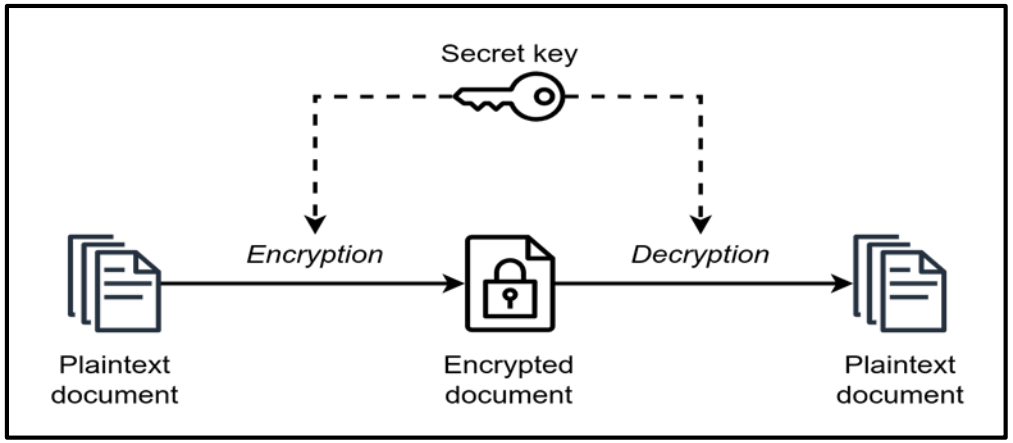
- Conventional / Secret key / Private-key / Single key.
- The sender and Receiver share a common key.
- All classical encryption algorithms are based on private keys.
- The keys may be identical, or there may be a simple transformation to go between the two keys.
- Symmetric-key encryption algorithms are usually better for bulk encryption.
- This contrasts with Asymmetric encryption.



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How Symmetric Encryption Works

- **Key Generation**
  - A secret key is generated using a secure random number generator.
  - This key is kept secret and is used for both encryption and decryption.
- **Encryption**
  - The secret key is used by an encryption algorithm to transform the data into ciphertext.
  - This ciphertext is unintelligible without the secret key.
- **Decryption**
  - The recipient uses the same secret key and a decryption algorithm.
  - The ciphertext is transformed back into the original plaintext.



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Advantages and Limitations of Symmetric Encryption

❖ Advantages

- **Speed and Efficiency**
  - Generally faster and more efficient than asymmetric.
  - Suitable for encrypting and decrypting large volumes of data.
- **Simplicity**
  - The encryption process is conceptually simpler.
  - Making it easier to implement and use.
- **Security**
  - The secret key is kept secret and is sufficiently long and random.
  - symmetric encryption can provide strong security.

❖ Limitations

- **Key Distribution**
  - Securely distributing can be challenging.
  - If the attacker intercepts, the security of the system is compromised.
- **Key Management**
  - Encrypted communication grows.
  - Managing and securely storing the secret keys becomes more complex.
- **Lack of Forward Secrecy**
  - Symmetric encryption does not provide forward secrecy.
  - All past and future communications encrypted with that key are also at risk.

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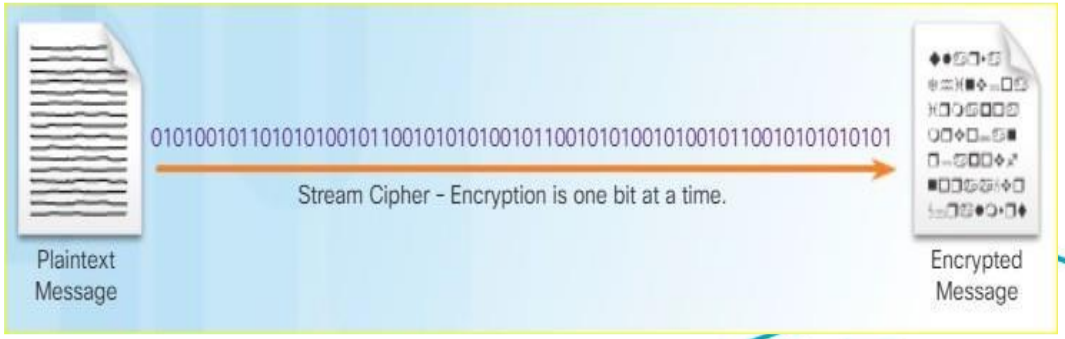
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Types of Symmetric Encryption

- **Block Ciphers**
  - Encrypt data in fixed-size blocks (e.g., 128, 192, or 256 bits)
  - Suitable for encrypting structured data like files and messages
  - Operated in various modes to handle different encryption scenarios
  - Common block ciphers include:
    - Advanced Encryption Standard (AES)
    - Data Encryption Standard (DES)
    - Triple Data Encryptions Standard (3DES)



- **Stream Ciphers**
  - Encrypt data bit by bit or byte by byte
  - Making them suitable for real-time communication and stream-based data
  - Often used in scenarios like encrypting network traffic
  - Common stream ciphers include:
    - Rivest Cipher 4 (RC4)
    - ChaCha20

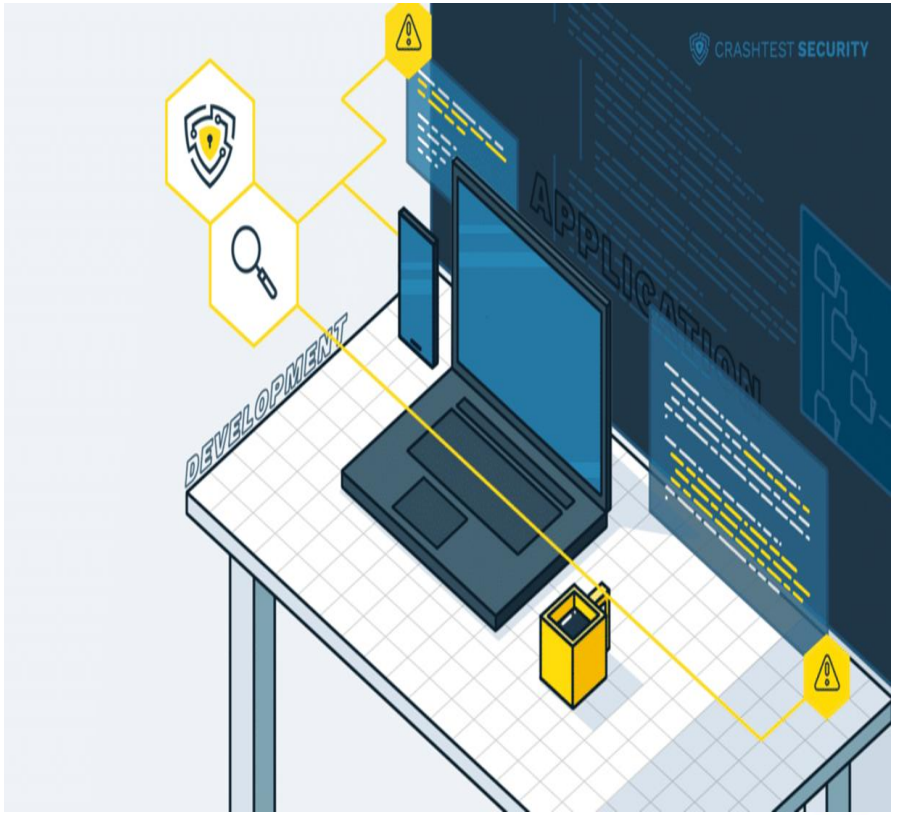


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Types of Symmetric Encryption (Contd.)

- **Synchronous vs. Asynchronous Ciphers**
  - Encryption and Decryption processes are synchronized using the same key and initialization vector (IV).
  - AES is an example of a synchronous block cipher.
  - Encryption and decryption may not be perfectly synchronized, and different keys or IVs may be used for different blocks of data.
  - Stream ciphers like RC4 and ChaCha20 can be considered asynchronous.
- **Mode of Operation**
  - Symmetric encryption algorithms can be used in various modes to adapt to different use cases.
  - In ECB, each block of plaintext is encrypted independently with the same key.
  - In CBC, each block is XORed with the ciphertext of the previous block before encryption.
  - In CTR, a block cipher is used in a counter mode to encrypt blocks of plaintext.
- **Block Size**
  - Algorithms come in various block sizes, such as 64 bits, 128 bits, or 256 bits
  - Block size can impact performance and security



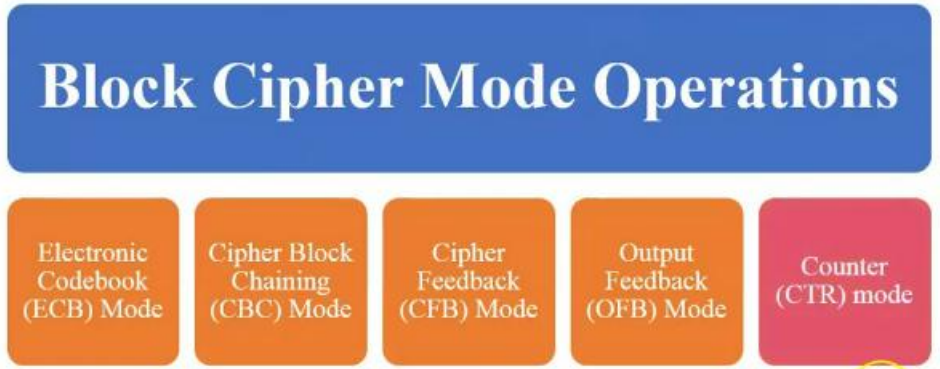
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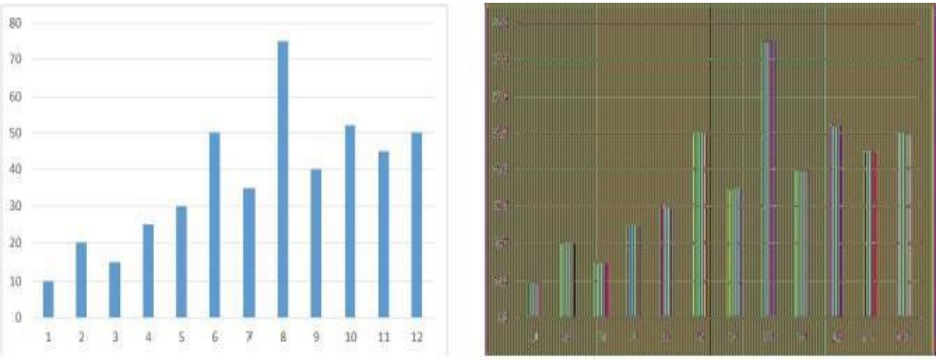
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Block Ciphers Mode

❖ Chaining Blocks

- Block ciphers take fixed-sized blocks mix them up and output a block of ciphertext.
- ***What happens when the input plaintext is bigger or smaller than this fixed block size?***
- Lots of different methods for padding out data that is smaller than the block size.
- Larger amounts of input data can be split into multiple blocks.
- Fixed-sized blocks are 64 or 128 bits at a time.
- Block ciphers are often used in different modes of operation.
- How the encryption process is applied to handle messages of arbitrary lengths.
- The problem is how the file is encrypted. If individual blocks are identical, then their encrypted blocks will be the same and can betray a pattern in the ciphertext.



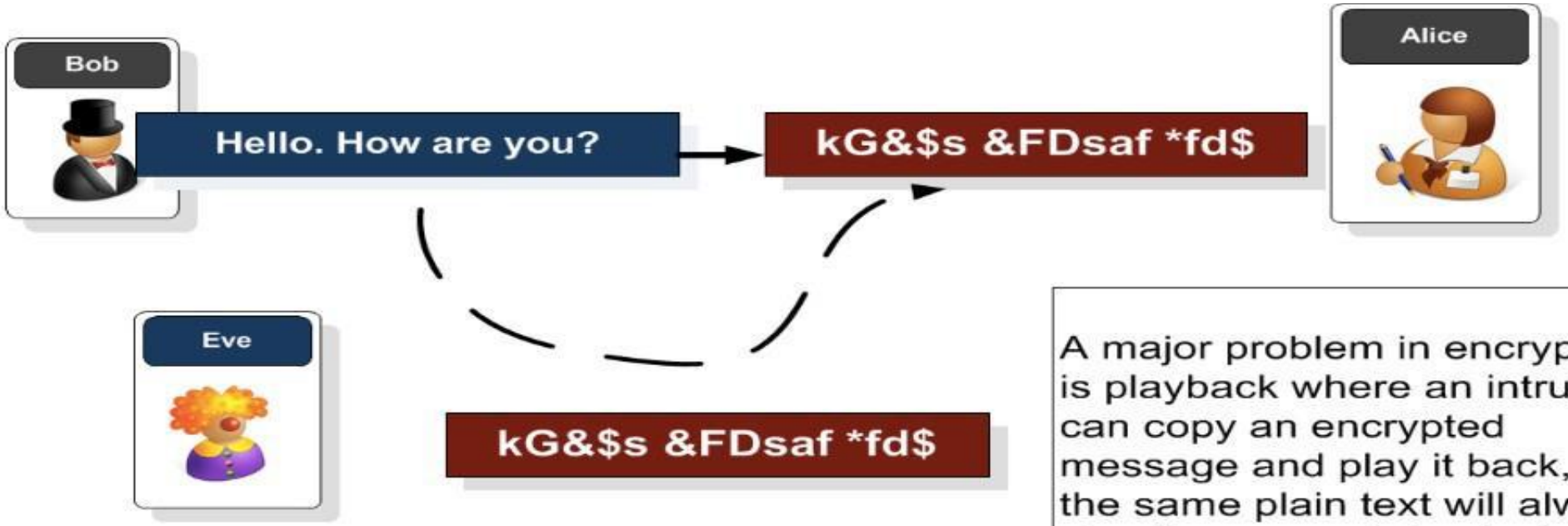


(a) The original image (pic_original.bmp) (b) The encrypted image (pic_encrypted.bmp

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Block Ciphers Mode Operation



A major problem in encryption is playback where an intruder can copy an encrypted message and play it back, as the same plain text will always give the same cipher text.

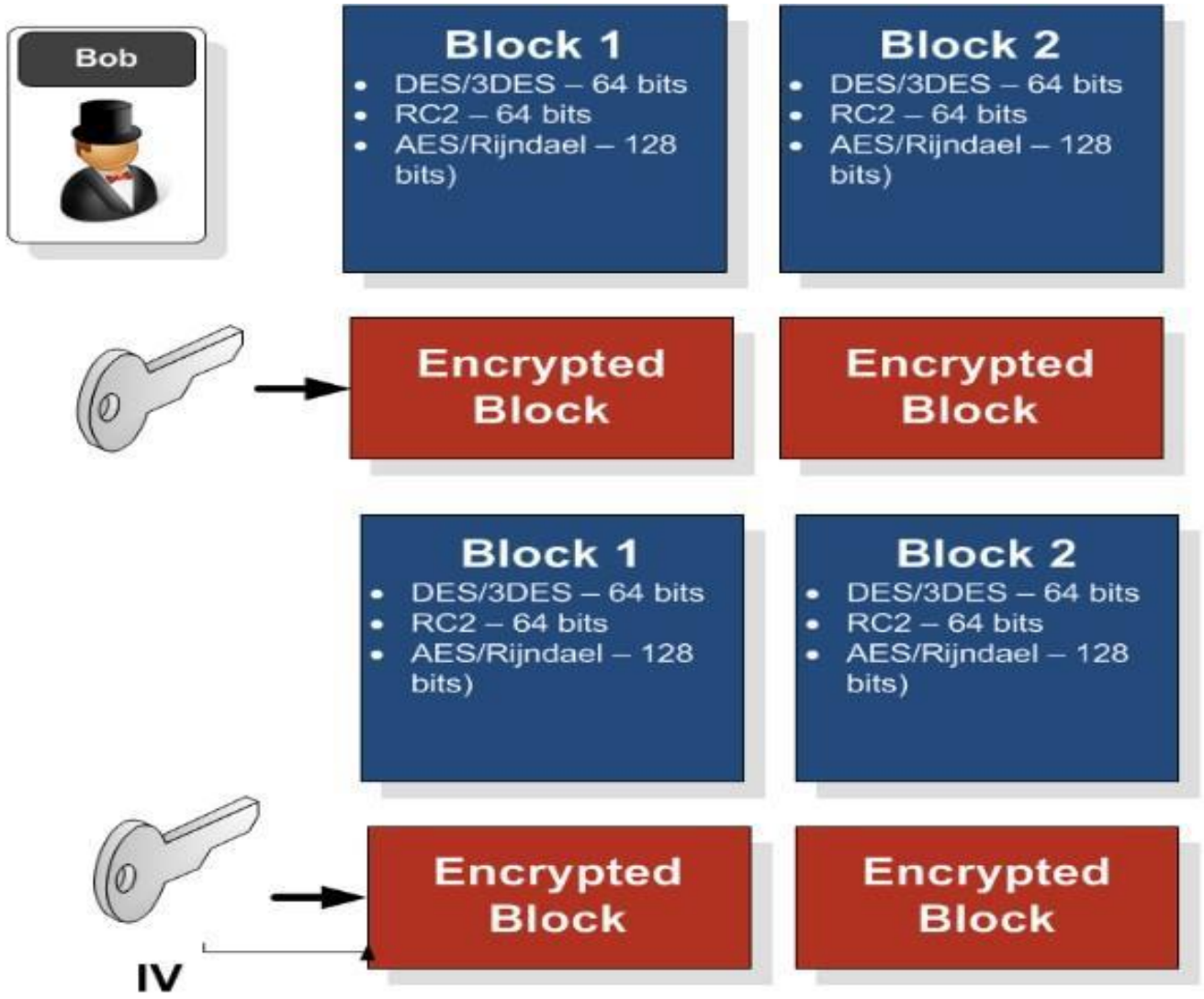


The solution is to add **salt** to the encryption key, as that it changes its operation from block-to-block (for block encryption) or data frame-to-data frame (for stream encryption)

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Block Ciphers Mode Operation (Contd.)



Electronic Code Book (ECB) method. This is weak, as the same cipher text appears for the same blocks.

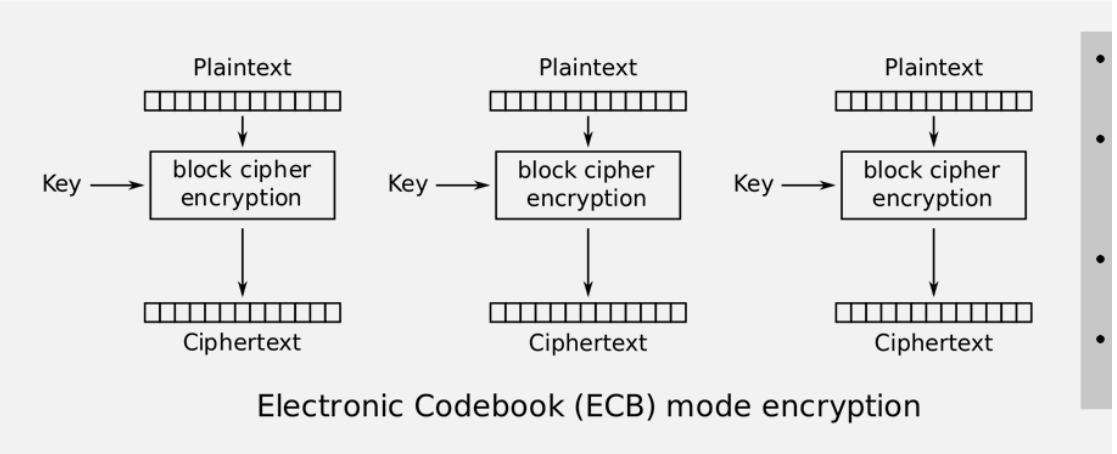
Hello → 5ghd%43f=
Hello → 5ghd%43f=

Adding salt. This is typically done with an IV (Initialisation Vector) which must be the same on both sides. In WEP, the IV is incremented for each data frame, so that the cipher text changes.

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Block Ciphers Mode

- **Electronic Codebook (ECB)**
  - The simplest and fastest chaining mode
  - Each block of plaintext independently uses the same encryption key
  - This means that identical plaintext blocks will produce identical ciphertext blocks
  - Not good on structured data, and susceptible to replay attacks



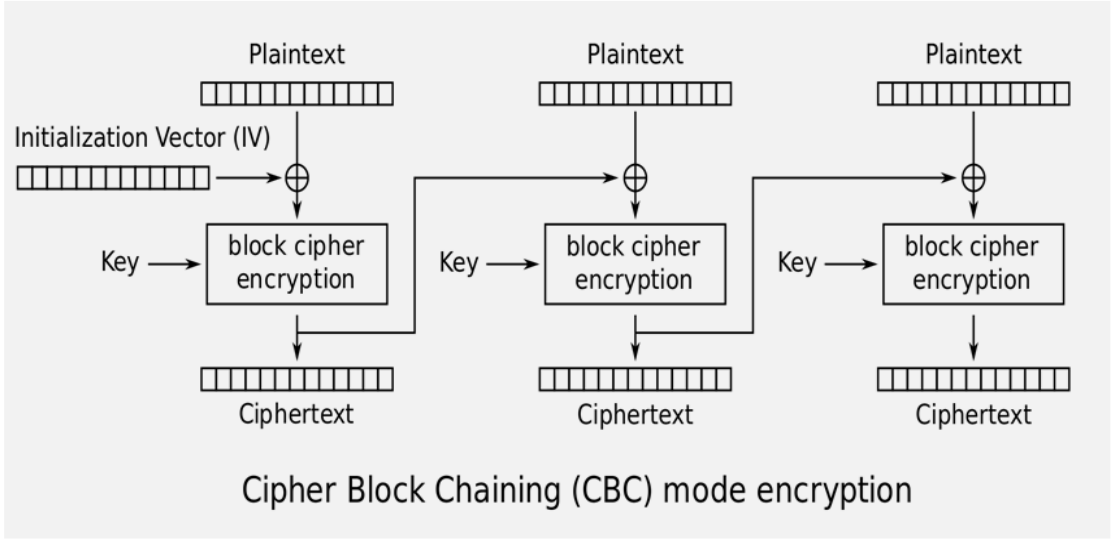
Electronic Codebook (ECB) mode encryption

- **For example**
  - If take "eeee

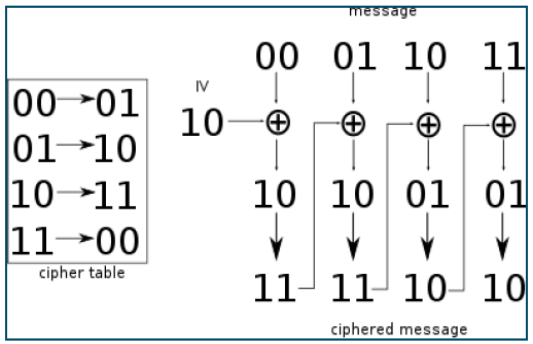
Block Ciphers Mode

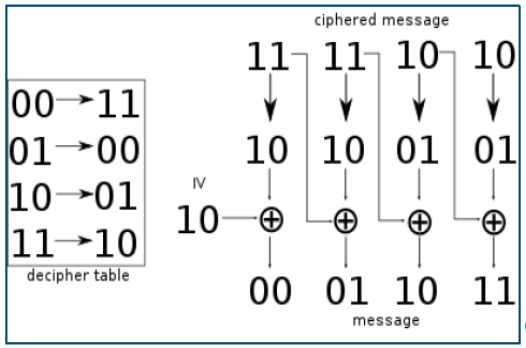
❖ Cipher Block Chaining (CBC)

- Each plaintext block is XORed with the ciphertext of the previous block before encryption
- CBC mode provides confidentiality and is resistant to identical plaintext attacks
- The main drawbacks are that encryption is sequential, decryption can be parallelized.
- Decrypting with the incorrect IV causes the first block of plaintext to be corrupt but subsequent plaintext blocks will be correct.
- Block 1 XOR IV:
  - Plaintext: "HELLO" (padded to 16 bytes as needed)
  - IV: [Random IV]
  - XOR Result: [XOR Result 1]
- Block 1 Decryption XOR IV:
  - Decryption Result: XOR Result 1
  - IV: [Random IV]
  - XOR Result: "HELLO" (padded to 16 bytes as needed)



Cipher Block Chaining (CBC) mode encryption



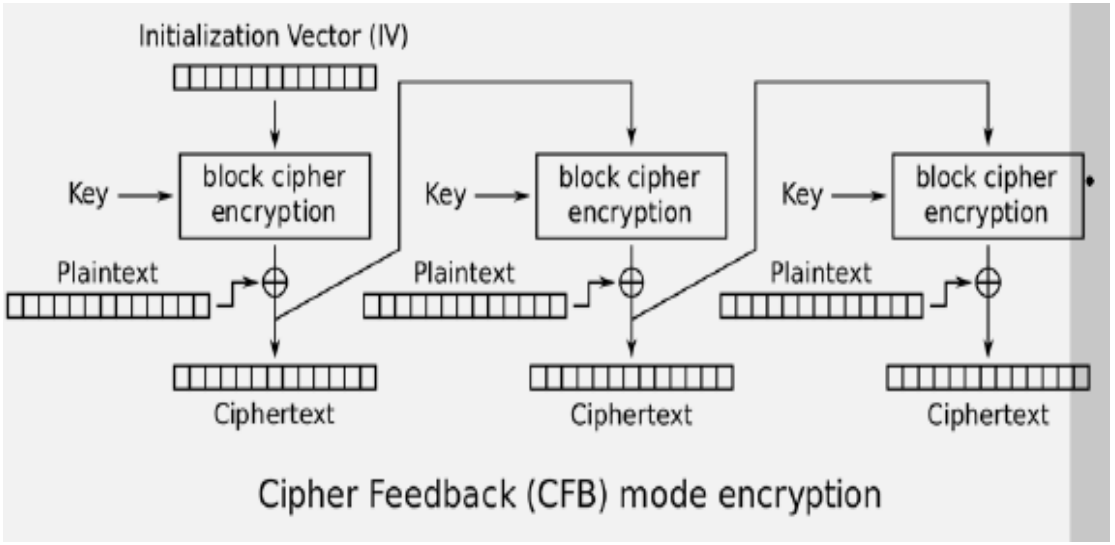


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Block Ciphers Mode

❖ Cipher Feedback (CFB)

- CFB mode turns a block cipher into a self-synchronizing stream cipher
- It operates on a block-by-block basis and allows encryption of arbitrary-length streams
- CFB mode provides confidentiality and error propagation
- When decrypting, a one-bit change in the ciphertext affects two plaintext blocks: a one-bit change in the corresponding plaintext block, and complete corruption of the following plaintext block. Later plaintext blocks are decrypted normally.
- Block 1 XOR IV:
  - Plaintext: "HELLO" (padded to 16 bytes as needed)
  - IV: [Random IV]
  - XOR Result: [XOR Result 1]
- Block 1 Decryption XOR IV:
  - Decryption Result: XOR Result 1
  - IV: [Random IV]
  - XOR Result: "HELLO" (padded to 16 bytes as needed)



Cipher Feedback (CFB) mode encryption

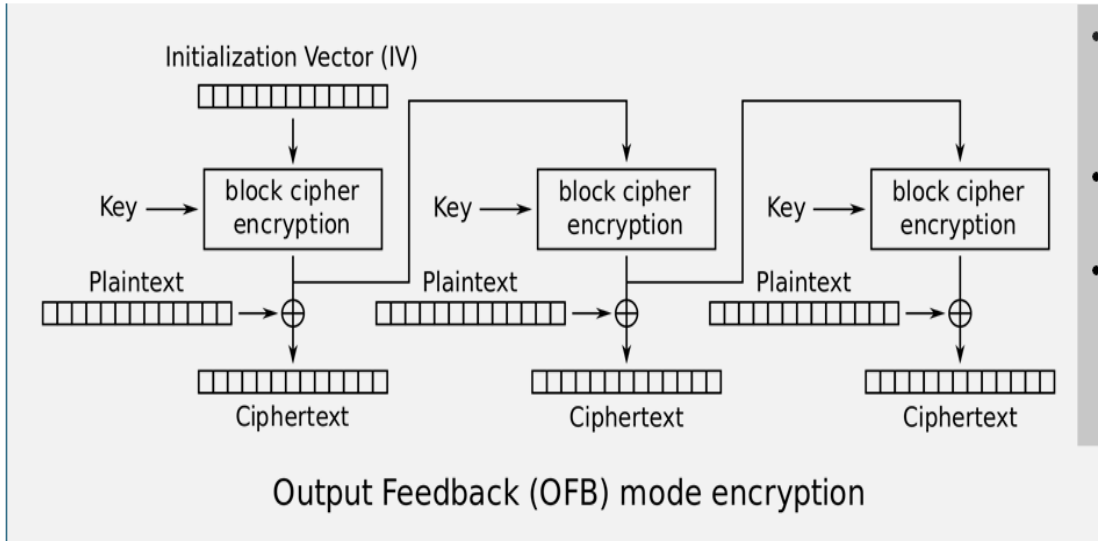
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Block Ciphers Mode

❖ Output Feedback (OFB)

- Makes a block cipher into a stream cipher
- OFB mode is primarily used for encryption rather than decryption, and cannot be performed in parallel.
- Allows provides confidentiality and error propagation
- Codes function normally even when applied before encryption
- Block 1 XOR Pseudorandom Stream:
  - Plaintext: "HELLO" (padded to 16 bytes as needed)
  - Pseudorandom Stream: [Pseudorandom stream]
  - XOR Result: [Ciphertext for Block 1]
- Block 1 Decryption XOR Pseudorandom Stream:
  - Ciphertext for Block 1: [Ciphertext for Block 1]
  - Pseudorandom Stream: [Pseudorandom stream]
  - XOR Result: "HELLO" (padded to 16 bytes as needed)

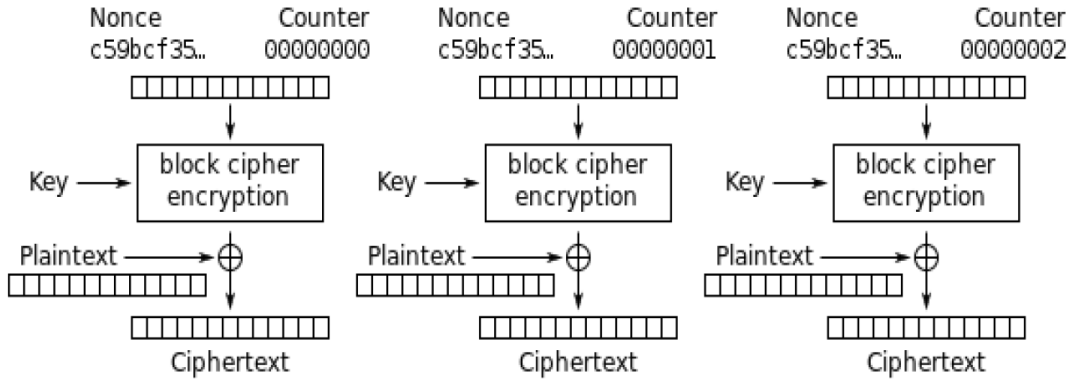


Output Feedback (OFB) mode encryption

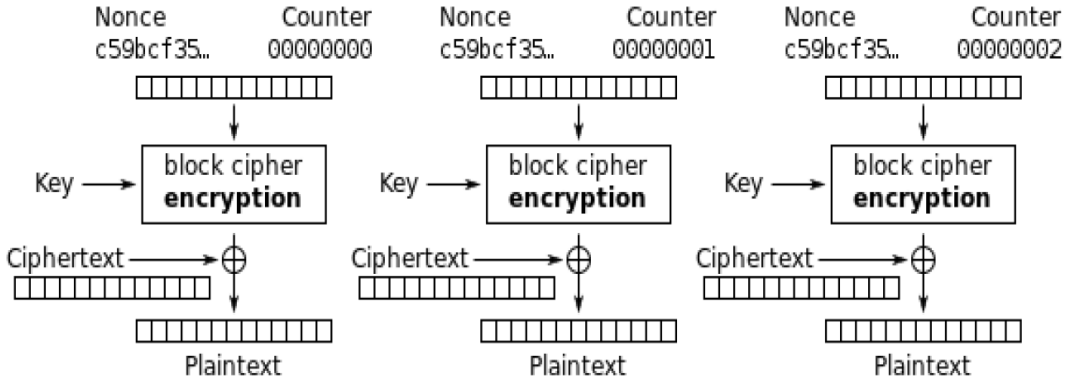
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Block Ciphers Mode

- **Counter (CTR) Mode**
  - It uses a counter to generate the key streams.
  - No key stream can be reused, hence the counter value for each block is prepended with a randomly generated value called *nonce* ***(number used once)***.
  - This nonce serves the same role as the IV does for the other encryption modes.
  - Both encryption and decryption can be parallelized.
  - The key stream in the CTR mode can be calculated in parallel.
- **Padding**
  - Block cipher encryption modes divide plaintext into blocks and the size of each block should match the cipher's block size.
  - No guarantee that the size of the last block matches the cipher's block size.
  - The last block of the plaintext needs padding i.e. before encryption, extra data needs to be added to the last block of the plaintext, so its size equals to the cipher's block size.
  - Padding schemes need to mark where the padding starts, so decryption can remove the padded data.
  - A commonly used padding scheme is Password-Based Cryptography Specification (PKCS#5).



Counter (CTR) mode encryption



Counter (CTR) mode decryption

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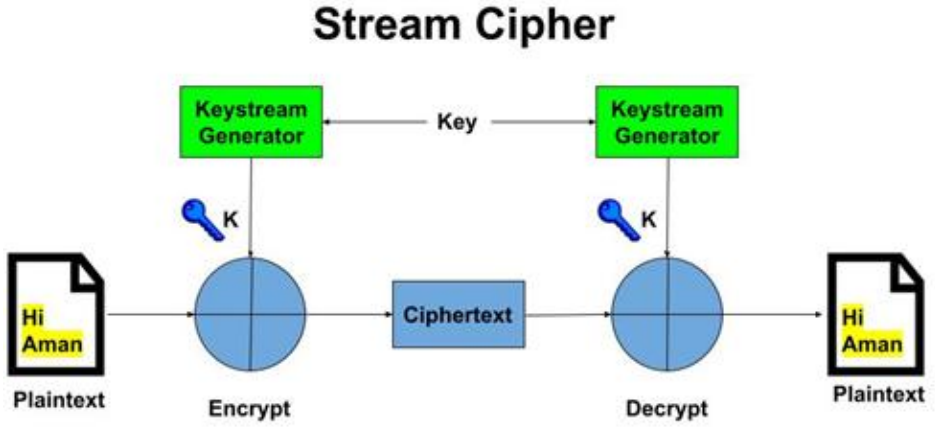
Stream Ciphers Mode

❖ Synchronous Stream Cipher

- ➢ The keystream is combined with the plaintext using a bitwise XOR operation.
- ➢ The same keystream is used for both encryption and decryption.
- ➢ The sender and receiver must be synchronized and use the same initial state for the stream cipher.
- ➢ Synchronous stream ciphers are simple and efficient.

❖ Self-Synchronizing Stream Cipher

- ➢ Unlike synchronous stream ciphers.
- ➢ Generate a keystream that depends on both the encryption key and previously generated bits of the keystream.
- ➢ This allows for resynchronization if some bits are lost or corrupted during transmission.
- ➢ However, resynchronization can be computationally expensive.



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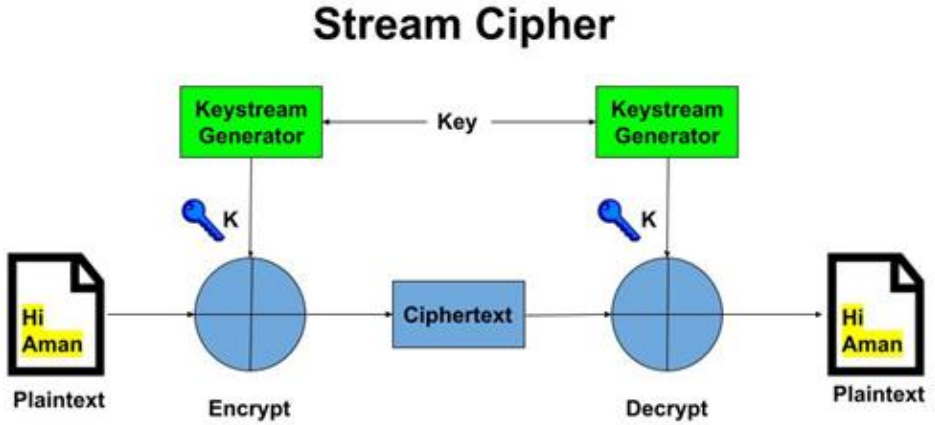
Stream Ciphers Mode (Contd.)

❖ Non-Self-Synchronizing Stream Cipher

- Non-self-synchronizing stream ciphers do not provide automatic resynchronization.
- That will affect any loss or corruption of bits in the keystream can result in decryption errors.
- They require careful handling and synchronization.

❖ Stream Cipher with Feedback

- Feedback mechanisms introduce dependencies between bits of the keystream.
- Makes it harder for an attacker to predict the keystream without knowing it.
- Some stream ciphers use feedback mechanisms e.g. Linear Feedback Shift Registers (LFSRs) or cellular automata.



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Stream Ciphers Mode (Contd.)

❖ Cipher Feedback (CFB) Mode

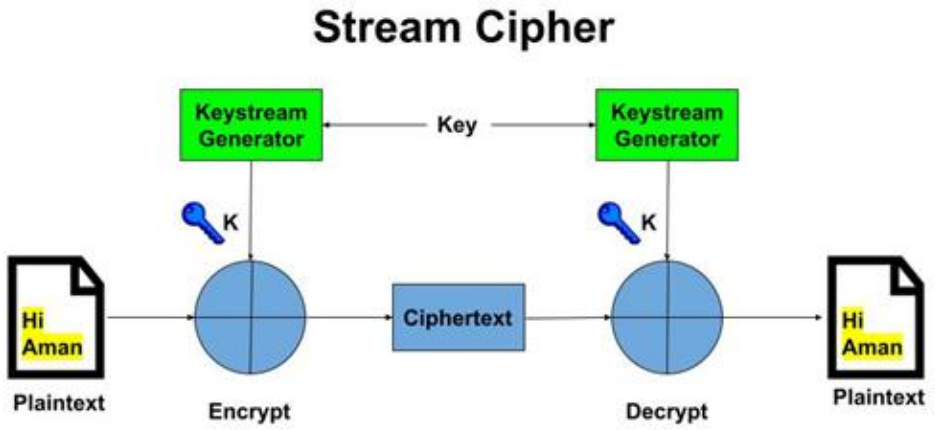
- CFB mode is to encrypt data in units larger than a single bit or byte.
- It turns a stream cipher into a block cipher-like encryption scheme.
- CFB mode allows for variable-length plaintext input, and it encrypts the plaintext block by block.

❖ Output Feedback (OFB) Mode

- OFB generates a keystream independently of the plaintext and then encrypts the plaintext with the keystream using XOR.
- OFB mode is particularly useful for encrypting data in a synchronous stream cipher when resynchronization is required.

❖ Counter (CTR) Mode

- CTR generates a keystream by encrypting a counter value with the stream cipher, which works as a block cipher-like scheme.
- CTR mode is parallelizable and provides good security properties.



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Symmetric Encryption Algorithm - DES

- **Data Encryption Standard (DES) -** This is an older legacy symmetric encryption algorithm.
- It can be used in **stream cipher mode** but usually operates in **block mode** by encrypting data in 64-bit block size.
- Every **8th key bit** is ignored in the DES algorithm so that the effective key size is **56 bits**.

DES Characteristics

Description	Data Encryption Standard
Timeline	Standardized 1976
Type of Algorithm	Symmetric
Key Size (in bits)	56 bits
Speed	Medium
Time to Crack (Assuming a computer could try 255 keys per second)	Days (6.4 days by the COPACABANA machine, a specialized cracking device)
Resource Consumption	Medium

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How DES Algorithm Working?

- **Plaintext message:** "8787878787878787".
- DES Key: **"0E329232EA6D0D73"**.
- **Ciphertext: "0000000000000000“**.
- A decrypted message with the same secret key: **"0E329232EA6D0D73“**.
- The result is the original plaintext: "8787878787878787".

- Plaintext message: "Your lips are smoother than Vaseline “, **38 Bytes (76 hexadecimal digits).**
- So this message must be **padded** with some extra bytes at the tail end for the encryption.
- Once the encrypted message has been decrypted, these extra bytes are **thrown away**.
- Here we will just add 0s at the end so that the total message is a **multiple of 8 bytes** (or 16 hexadecimal digits, or 64 bits).

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Symmetric Encryption Algorithm - 3DES

- **3DES (Triple DES) -** This is a newer version of DES where it repeats the DES algorithm process three times.
- It is more computationally taxing than DES.
- The basic algorithm has been well-tested in the field for more than 35 years and is therefore considered very trustworthy when implemented using very short key lifetimes.

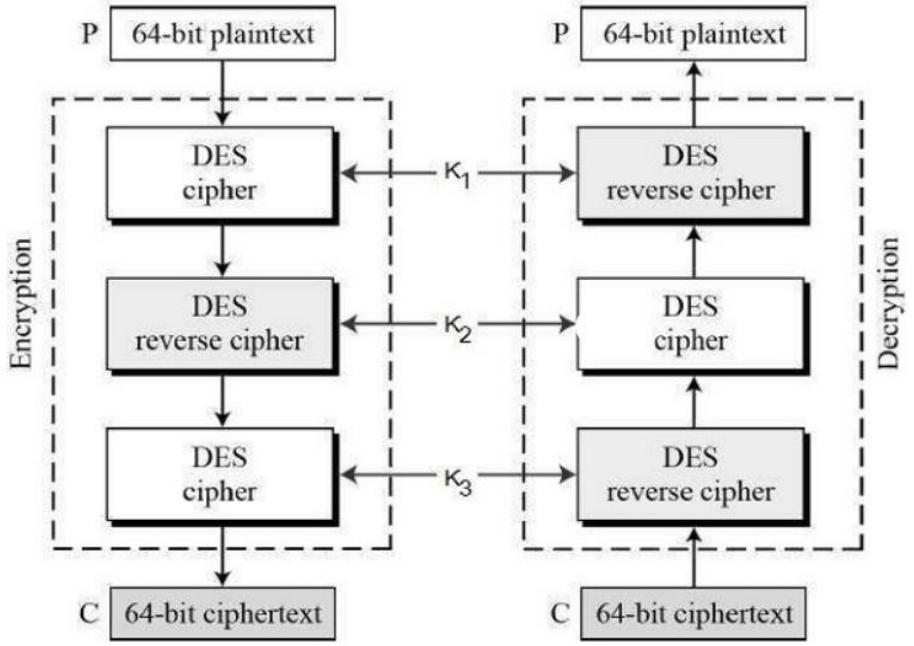
3DES Characteristics	
Description	Triple Data Encryption Standard
Timeline	Standardized 1977
Type of Algorithm	Symmetric
Key Size (in bits)	112 and 168 bits
Speed	Low
Time to Crack (Assuming a computer could try 255 keys per second)	4.6 billion years with current technology
Resource Consumption	Medium

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How 3DES Algorithm Working?

- Encrypt the plaintext blocks using a **single DES** with key K_1.
- Now decrypt the **output** of step 1 using a single DES with key K_2.
- Finally, encrypt the output of **step 2** using a single DES with key K_3.
- The **output of step 3** is the ciphertext.
- Decryption of a ciphertext is a **reverse process**. User first decrypt using K_3, then encrypt with K_2, and finally decrypt with K_1.



- **NOTE:** Due to this design of Triple DES as an encrypt–decrypt–encrypt process, it is possible to use a 3TDES (hardware) implementation for single DES by setting K1, K2, and K3 to be the same value. This provides backwards compatibility with DES.

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Symmetric Encryption Algorithm - SEAL

- **Software Optimized Encryption Algorithm (SEAL) -** is a fast alternative symmetric encryption algorithm to DES, 3DES, and AES.
- It is a **stream cipher** that uses a 160-bit encryption key.
- SEAL has a **lower impact** on the CPU compared to other software-based algorithms but is still considered unproven.

SEAL Characteristics	
Description	Software-Optimized Encryption Algorithm
Timeline	First published in 1994, current version is 3.0 (1997)
Type of Algorithm	Symmetric
Key Size (in bits)	160
Speed	High
Time to Crack (Assuming a computer could try 255 keys per second)	Unknown but considered very safe
Resource Consumption	Low

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Symmetric Encryption Algorithm – Rivest Ciphers (RC)

- **Rivest ciphers (RC)** series algorithms (includes RC2, RC4, RC5, and RC6).
- This was developed by Ron Rivest. Several variations have been developed, but RC4 is the most prevalent in use.
- RC4 is a fast and efficient encryption algorithm that is suitable for low-power devices and applications that require high-speed data transmission.
- RC4 is a stream cipher and is used to secure web traffic in SSL and TLS.
- RC4 has several vulnerabilities, including a bias in the first few bytes of the keystream, which can be exploited to recover the key.

RC Algorithms	Timeline	Type of Algorithm	Key Size in Bits
RC2	1987	Block cipher	40 and 64
RC4	1987	Stream cipher	1 to 256
RC5	1994	Block cipher	0 to 2048
RC6	1998	Block cipher	128, 192, or 256

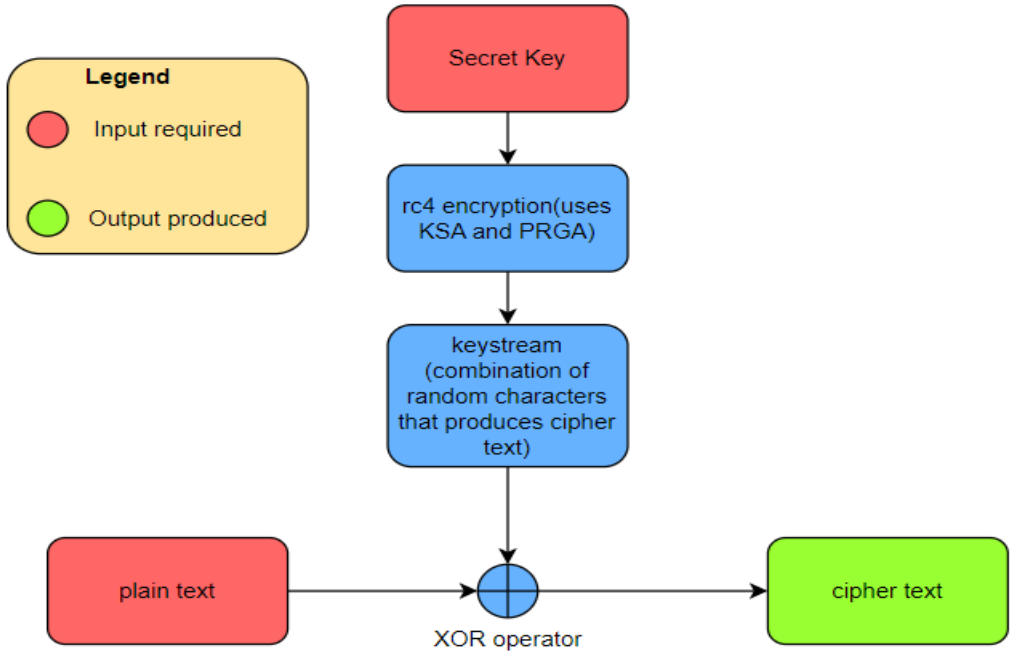


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How Rivest Ciphers (RC4) Algorithm Working

- The **RC4 (Rivest Cipher 4) encryption algorithm** is a symmetric key encryption algorithm that encrypts plain text in small chunks producing multiple stream cipher.
- RC4 encryption algorithm comprises two components—KSA (Key scheduling algorithm) and PRGA (Pseudo random generation algorithm).
- These two algorithms together help the rc4 algorithm to produce the stream cipher.



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How Rivest Ciphers (RC4) Algorithm Working (Contd.)

Encryption

- The user inputs plain text and a secret key.
- The encryption engine generates the keystream by using KSA and PRGA Algorithms for the secret key entered.
- The generated keystream is XORed with plain text. Since RC4 is a stream cipher, XORing is done byte by byte and encrypted text is produced.
- This encrypted text is now sent to the intended receiver in encrypted form.

Plain Text: 10011001

Keystream: 11000011

Cipher Text: 01011010

Decryption

- For decryption, ciphertext and the same keystream are required which was used for encryption.
- The ciphertext and the keystream produce plain text using XOR Operation.
- The ciphertext is XOR'ed with keystream bit by bit to produce PlainText.

Cipher Text: 01011010

Keystream: 11000011

Plain Text: 10011001

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Symmetric Encryption Algorithm - AES

- **Advanced Encryption Standard (AES) -** It offers nine combinations of key and block length by using a variable key length of 128 192 or 256 bits key to encrypt data blocks that are 128, 192, or 256 bits long. AES is a secure and more efficient algorithm than 3DES.

AES Characteristics

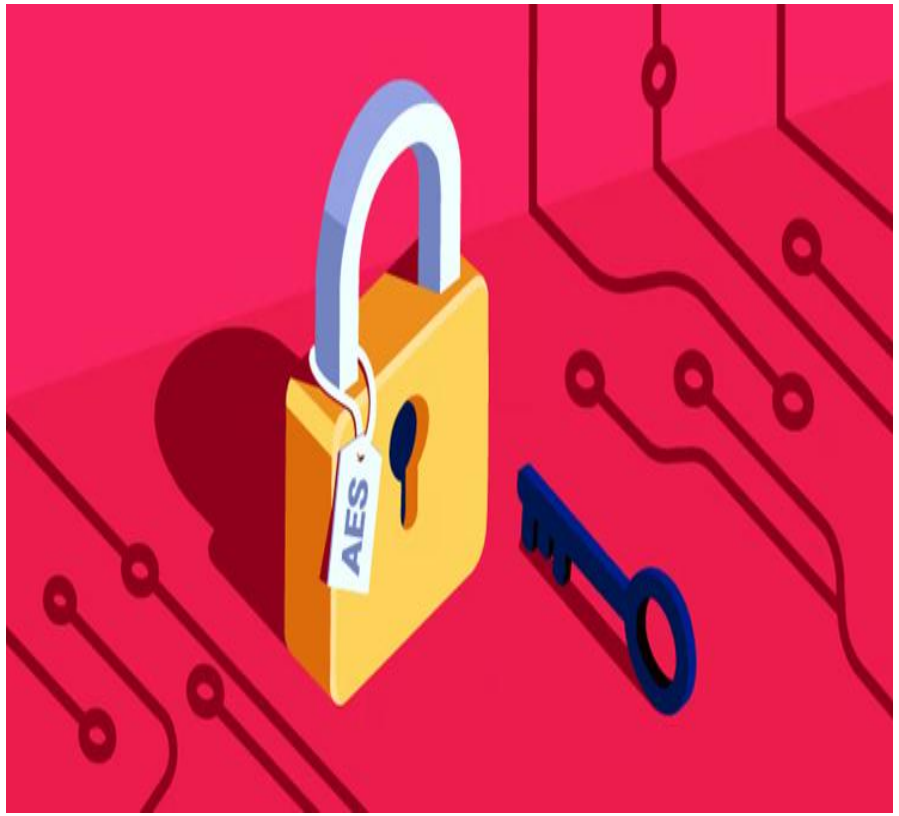
Description	Advanced Encryption Standard
Timeline	Official Standard since 2001
Type of Algorithm	Symmetric
Key Size (in bits)	128, 192, and 256
Speed	High
Time to Crack (Assuming a computer could try 255 keys per second)	149 trillion years
Resource Consumption	Low

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Advanced Encryption Standard

- AES was established by NIST in 2001, replacing the aging DES.
- AES is implemented in software and hardware throughout the world to encrypt sensitive data.
- It is essential for government computer security, cybersecurity, and electronic data protection.
- AES is a symmetric block cipher algorithm, and includes three block ciphers:
  - AES-128 uses 128-bit (10 rounds).
  - AES-192 uses 192-bit (12 rounds).
  - AES-256 uses 256-bit (14 rounds).
- Regardless of the key size, the input block size is always 16 bytes (a 4 * 4 matrix)
  - This is the initial starting state.

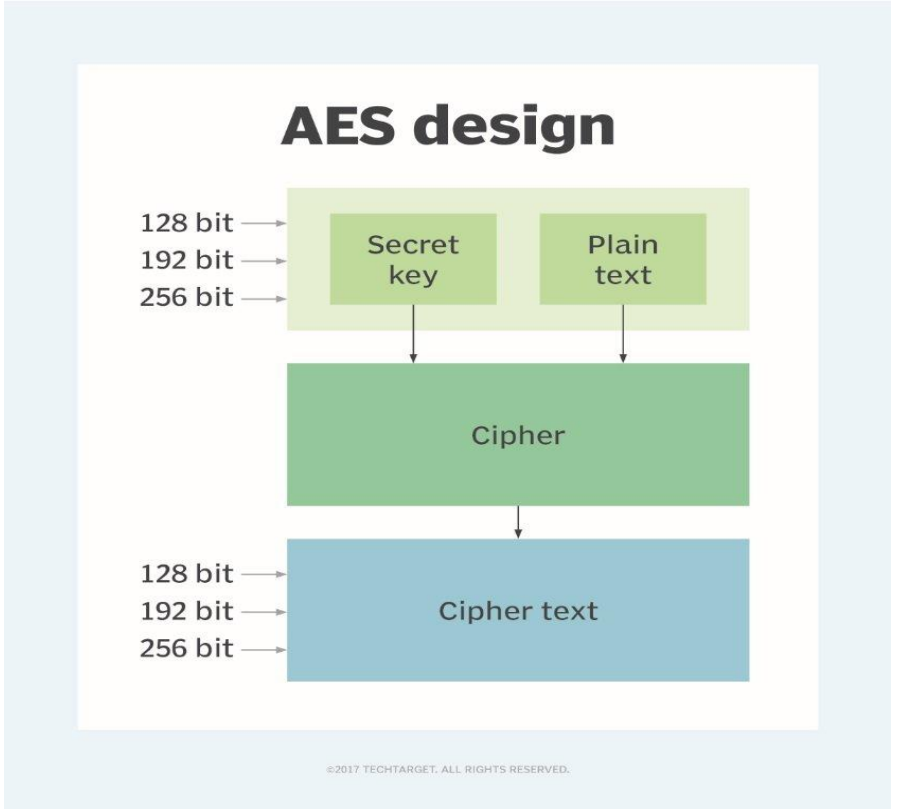


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AES Phases

- Key Expansion or Key Schedule
  - The original secret key is expanded into a set of round keys.
  - The key expansion algorithm varies based on the key size (128-bit, 192-bit, or 256-bit).
- Initial Round (AddRoundKey)
  - The initial round key is added, and the round key for the first round is XORed with each block (128 bits for AES-128).
- Rounds (SubBytes, ShiftRows, MixColumns, AddRoundKey)
  - AES operates on multiple rounds (10 rounds for AES-128, 12 rounds for AES-192, and 14 rounds for AES-256).
- Final Round (SubBytes, ShiftRows, AddRoundKey)
  - Similar to the regular rounds, but omits the MixColumns operation to avoid over-complication.
- Output
  - After the final round, the resulting data is the ciphertext (for encryption) or plaintext (for decryption).

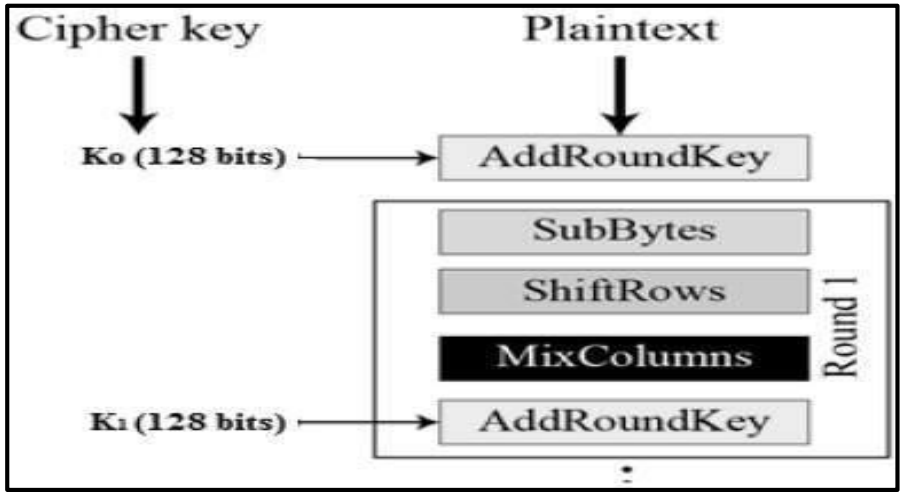


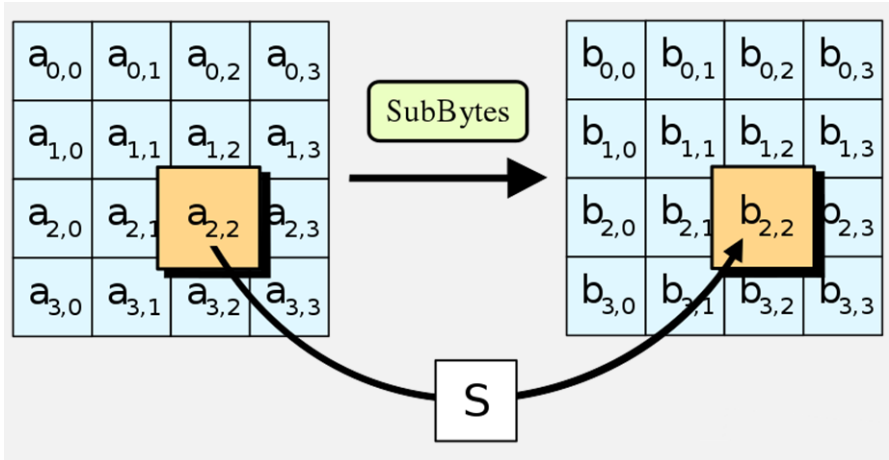
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Elements of each AES Round

- SubBytes
  - Each byte of the state is substituted for a different byte according to a lookup table ("S-box").
- ShiftRows
  - The last three rows of the state matrix are transposed.
  - Shifting the rows of the block to provide diffusion. Each row is shifted by a varying number of bytes.
- MixColumns
  - Matrix multiplication is performed on the columns of the state, combining the four bytes in each column.
  - This stage is skipped in the final round.
- AddRoundKey
  - XORing the block with the corresponding round key for that round.





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Elements of each AES Round (Contd.)

- SubBytes Input State

  19 A0 9A E9
  3D F4 C6 F8
  E3 E2 8D 48
  BE 2B 2A 08

- After SubBytes (using S-box lookup for each byte)

  D4 E0 B8 1E
  27 BF B4 41
  11 98 5D 52
  AE F1 E5 30

Row \Column	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

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Elements of each AES Round (Contd.)

- ShiftRows in AES

Row	Shift
0	0 bytes
1	1 byte
2	2 bytes
3	3 bytes

- Before Shifting

D4 E0 B8 1E
27 BF B4 41
11 98 5D 52
AE F1 E5 30

- After Shifting

D4 E0 B8 1E
BF B4 41 27
5D 52 11 98
30 AE F1 E5

MixCloulumns

Each column of the state is multiplied by a **fixed 4×4 matrix** (called the MixColumns matrix):

$$\text{NewColumn} = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \times \begin{bmatrix} D4 \\ BF \\ 5D \\ 30 \end{bmatrix}$$

In AES:

- Multiplying by `1` = same value
- Multiplying by `2` = left shift by 1 bit (and XOR with `0x1B` if overflow occurs)
- Multiplying by `3` = (2 × value) XOR (1 × value)

Let's compute the first byte of the new column:

$$b_0' = (02 \times D4) \oplus (03 \times BF) \oplus (01 \times 5D) \oplus (01 \times 30)$$

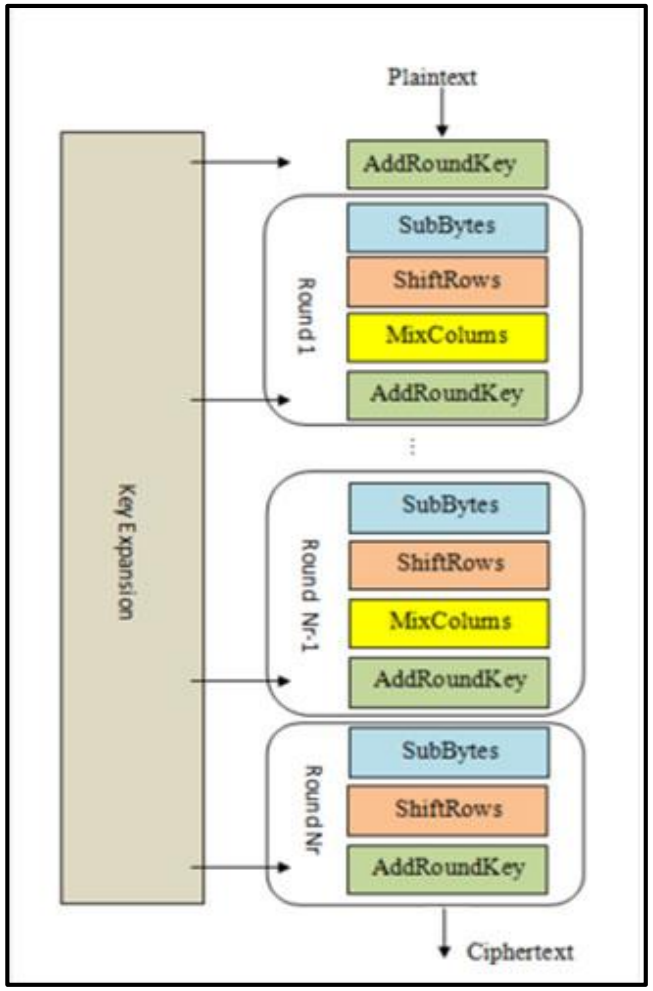
New Byte	Formula	Result
b_0'	(2×D4) ⊕ (3×BF) ⊕ (1×5D) ⊕ (1×30)	04
b_1'	(1×D4) ⊕ (2×BF) ⊕ (3×5D) ⊕ (1×30)	66
b_2'	(1×D4) ⊕ (1×BF) ⊕ (2×5D) ⊕ (3×30)	81
b_3'	(3×D4) ⊕ (1×BF) ⊕ (1×5D) ⊕ (2×30)	E5

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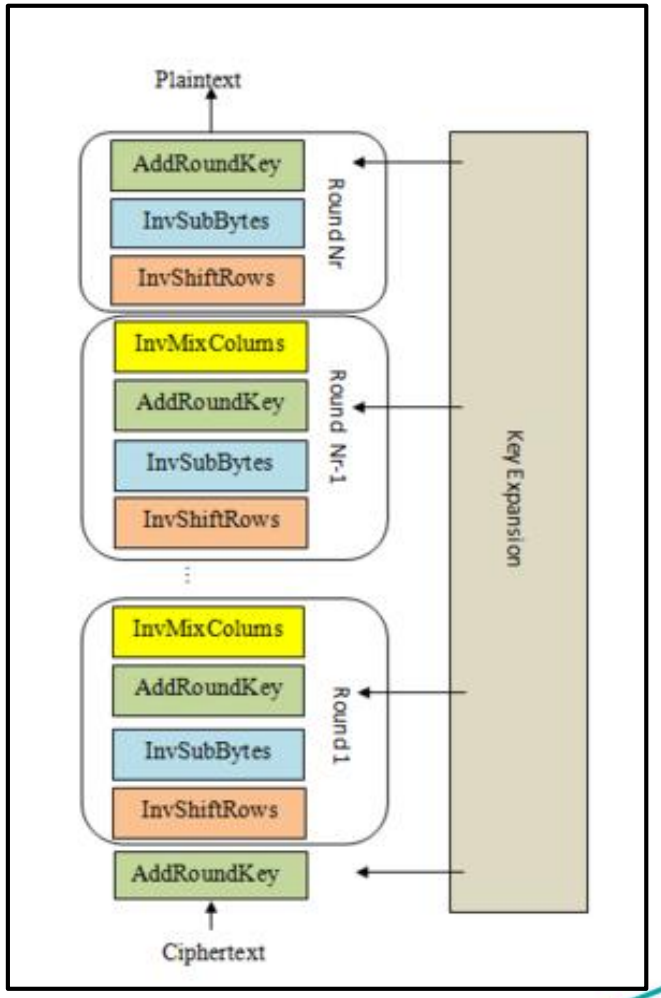
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AES Structure

AES Encryption



AES Decryption



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Short Quiz

- **Which mode of operation for a block cipher has the characteristic that each possible block of plaintext has a defined corresponding ciphertext value and vice versa?**

A. Footprinting B. Hash function C. Watermark D. Electronic code book

- Electronic code book.

- **Which cryptography approach is most often used by today's internet businesses and users?**

A. Public key infrastructure B. Output feedback C. Encrypting file system D. Single sign=on

- Public key infrastructure.

- **What is a trial-and-error method used to decode encrypted data through exhaustive effort rather than employing intellectual strategies?**

A. Chaffing and winnowing B. Cryptanalysis C. Serendipity D. Brute-force cracking

- Brute-force cracking

- **What is the name of the encryption/decryption key known only to the party or parties that exchange secret messages?**

A. E-signature B. Digital certificate C. Private key D. Security token

- Private key

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Practical Exercise

- **Week 4 - 9th Oct**
  - CryptoHack.org website - Complete Introduction and General Challenges **(Encoding and XOR challenges)**. (Optional)
  - Read **Chapter 2, Secret Key Encryption** Cryptography by William Buchanan.
  - Attempt **Lab 5 - Symmetric Key Encryption, Lab 9 - Key Exchange,** and **Lab 10 - Public Key** Worksheets during the lab session.
  - Lab 6 - **Encrypted Image (Challenge Lab)** due on 18th October.
  - **Test 1** on **23rd October** during the lab session

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